EX.NO: 1a

PROGRAM

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:background="#d0d69f">
    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/uname"
        android:textColor="#ff0000"
        android:textSize="30sp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
    <EditText
        android:id="@+id/editText1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:inputType="textPersonName"
        app:layout_constraintStart_toEndOf="@+id/textView1"
        app:layout_constraintTop_toTopOf="@+id/textView1"
        app:layout_constraintEnd_toEndOf="parent" />
    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/pword"
        android:textColor="#ffff00"
        android:textSize="30sp"
        android:layout_marginTop="20dp"
        app:layout_constraintStart_toStartOf="@+id/textView1"
        app:layout_constraintTop_toBottomOf="@+id/textView1" />
    <EditText
```

```
        android:id="@+id/editText2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:inputType="textPassword"
        app:layout_constraintStart_toEndOf="@+id/textView2"
        app:layout_constraintTop_toTopOf="@+id/textView2"
        app:layout_constraintEnd_toEndOf="parent" />
    <RadioButton
        android:id="@+id/radioButton1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/cse"
        android:layout_marginTop="20dp"
        app:layout_constraintStart_toStartOf="@+id/textView2"
        app:layout_constraintTop_toBottomOf="@+id/editText2" />
    <RadioButton
        android:id="@+id/radioButton2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/it"
        app:layout_constraintStart_toStartOf="@+id/radioButton1"
        app:layout_constraintTop_toBottomOf="@+id/radioButton1" />
    <RadioButton
        android:id="@+id/radioButton3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/ece"
        app:layout_constraintStart_toStartOf="@+id/radioButton2"
        app:layout_constraintTop_toBottomOf="@+id/radioButton2" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/submit"
        android:textColor="#FFFFFF"
        android:background="#4CAF50"
        android:paddingLeft="20dp"
        android:paddingRight="20dp"
        android:paddingTop="10dp"
        android:paddingBottom="10dp"
        android:layout_marginTop="30dp"
```

```
        app:layout_constraintTop_toBottomOf="@+id/radioButton3"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

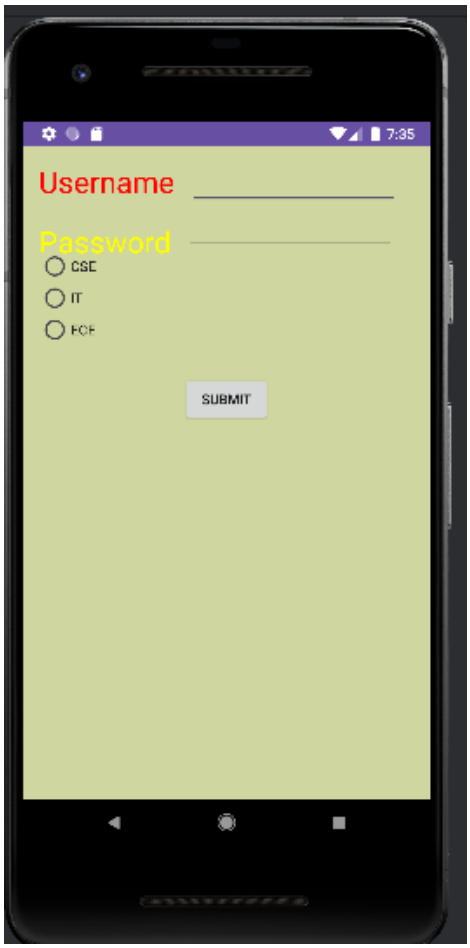
MainActivity.java

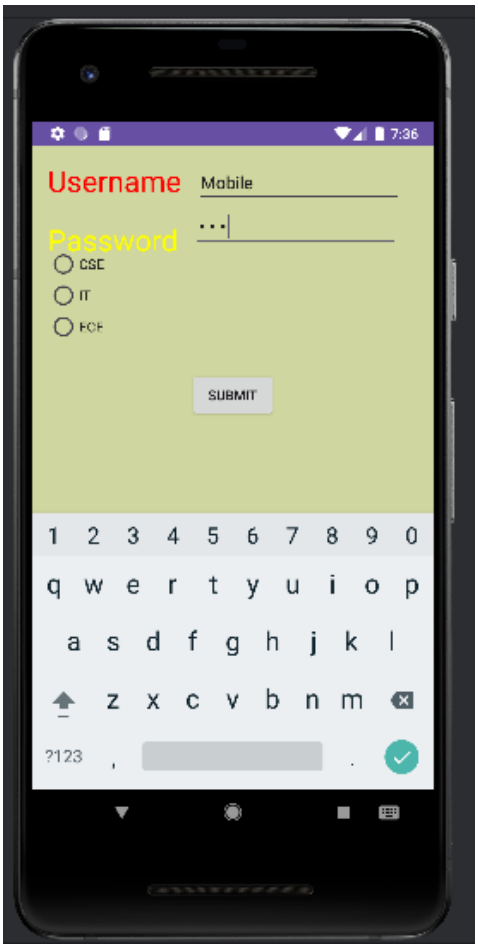
```
package com.example.experiment1;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.content.res.ResourcesCompat;
import android.graphics.Typeface;
import android.os.Bundle;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button bt = findViewById(R.id.button1);
        // Load font from res/font
        Typeface font = ResourcesCompat.getFont(this, R.font.roboto_slab);
        // Apply font to Button
        bt.setTypeface(font);
    }
}
```

strings.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">Exercise1</string>
<string name="uname">Username</string>
<string name="pword">Password</string>
<string name="cse">CSE</string>
<string name="it">IT</string>
<string name="ece">ECE</string>
<string name="submit">SUBMIT</string>
</resources>
```

OUTPUT


7:35
Username
Password
CSE
IT
ECE
SUBMIT


7:36
Username
Mobile
Password
CSE
IT
ECE
SUBMIT

EX.NO:1b

PROGRAM

activity_main.xml

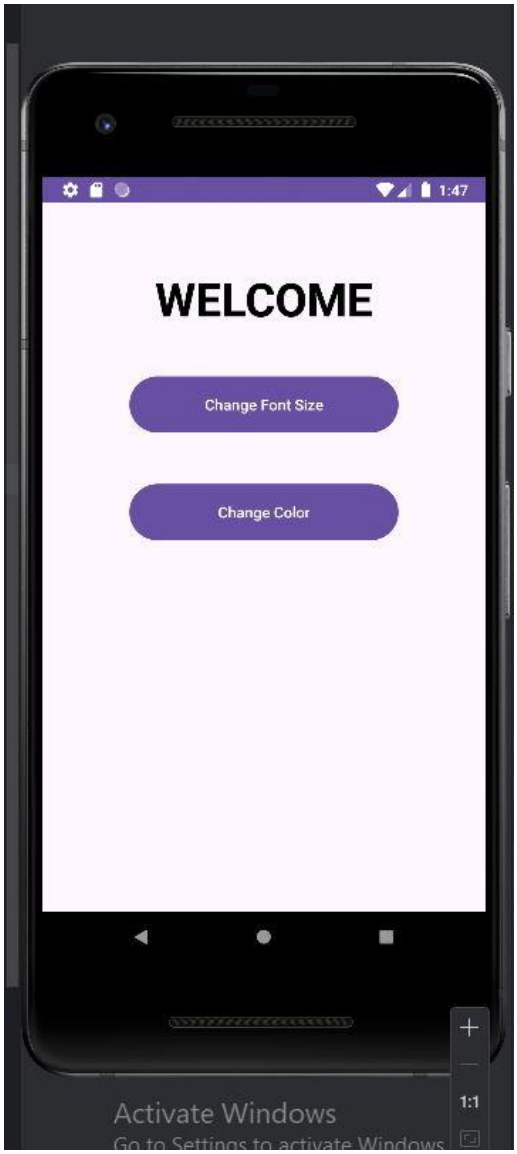
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">
    <TextView
        android:id="@+id/tvWelcome"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="WELCOME"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="40dp"
        android:textColor="#000000"/>
    <Button
        android:id="@+id/btnFont"
        android:layout_width="250dp"
        android:layout_height="60dp"
        android:text="Change Font Size"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="40dp"
        android:layout_below="@id/tvWelcome" />
    <Button
        android:id="@+id/btnColor"
        android:layout_width="250dp"
        android:layout_height="60dp"
        android:text="Change Color"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="40dp"
        android:layout_below="@id/btnFont" />
</RelativeLayout>
```

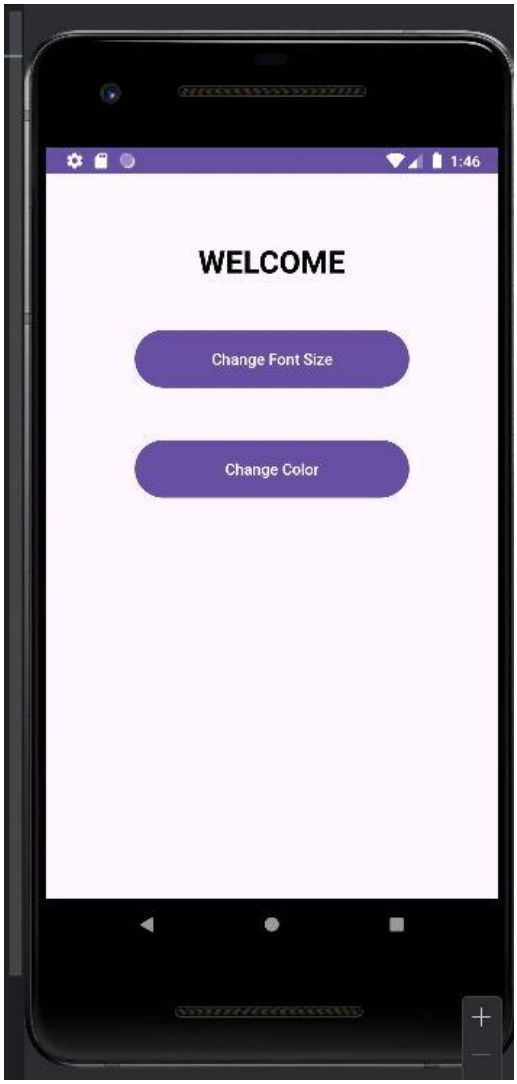
MainActivity.java

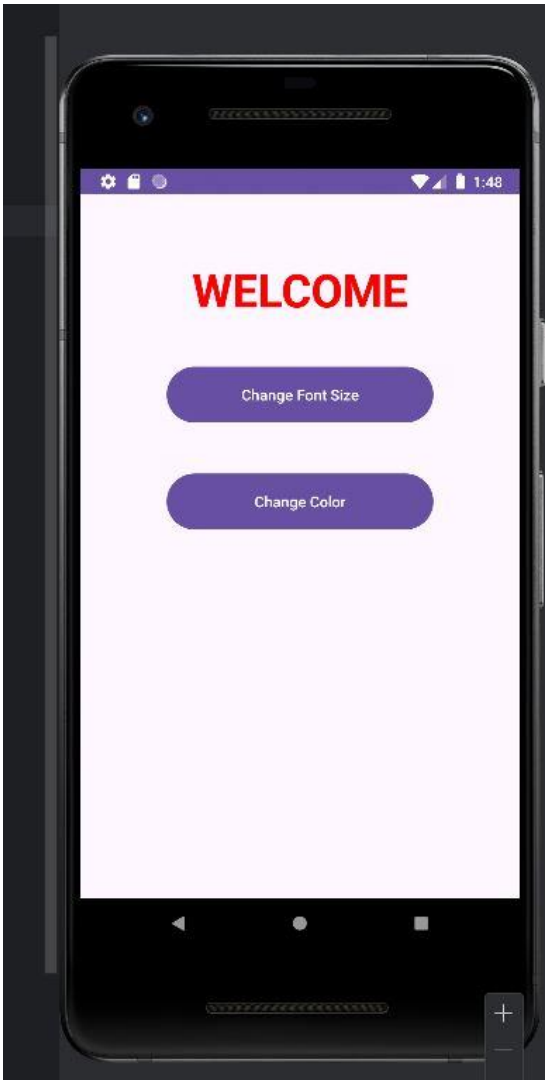
```
package com.example.exp2a;
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    TextView tvWelcome;
    Button btnFont, btnColor;
    int fontSize = 28;  // initial font size
    boolean isColorChanged = false;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        tvWelcome = findViewById(R.id.tvWelcome);
        btnFont = findViewById(R.id.btnFont);
        btnColor = findViewById(R.id.btnColor);
        // Change Font Size
        btnFont.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                fontSize += 5;     // increase font size
                if (fontSize > 60) fontSize = 28;  // reset if too large
                tvWelcome.setTextSize(fontSize);
            }
        });
        // Change Text Color
        btnColor.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                if (!isColorChanged) {
                    tvWelcome.setTextColor(Color.RED);
                } else {
                    tvWelcome.setTextColor(Color.BLACK); // toggle back
                }
                isColorChanged = !isColorChanged;
            }
        });  }}
```

OUTPUT:


1:47
WELCOME
Change Font Size
Change Color
Activate Windows
1:1


1:46
WELCOME
Change Font Size
Change Color


1:48
WELCOME
Change Font Size
Change Color

EX.NO: 2a

PROGRAM

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <!-- TITLE -->
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Enter Some Text"
        android:textSize="22sp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <!-- EDIT TEXT -->
    <EditText
        android:id="@+id/editText1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Type here"
        android:textSize="18sp"
        android:padding="10dp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@id/textView"
        android:layout_marginTop="10dp"/>

    <!-- SIZE LABEL -->
    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Change Text Size"
        android:textSize="18sp"
        android:layout_marginTop="20dp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/editText1"/>

    <!-- BUTTON 1 : 25SP -->
    <Button
        android:id="@+id/button1"
        android:layout_width="90dp"
        android:layout_height="wrap_content"
        android:text="25SP"
        android:background="#D3D3D3"
        android:layout_marginTop="10dp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/textView2"/>

    <!-- BUTTON 2 : 35SP -->
    <Button
        android:id="@+id/button2"
        android:layout_width="90dp"
        android:layout_height="wrap_content"
        android:text="35SP"
        android:background="#D3D3D3"
        app:layout_constraintStart_toEndOf="@id/button1"
        app:layout_constraintTop_toTopOf="@id/button1"
        android:layout_marginStart="15dp"/>

    <!-- BUTTON 3 : 45SP -->
    <Button
        android:id="@+id/button3"
        android:layout_width="90dp"
        android:layout_height="wrap_content"
        android:text="45SP"
        android:background="#D3D3D3"
        app:layout_constraintStart_toEndOf="@id/button2"
        app:layout_constraintTop_toTopOf="@id/button1"
        android:layout_marginStart="15dp"/>

    <!-- COLOR BUTTON 1 (RED) -->
    <Button
        android:id="@+id/button4"
        android:layout_width="90dp"
```

```
        android:layout_height="50dp"
        android:background="#FF0000"
        android:layout_marginTop="20dp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/button1" />

    <!-- COLOR BUTTON 2 (GREEN) -->
    <Button
        android:id="@+id/button5"
        android:layout_width="90dp"
        android:layout_height="50dp"
        android:background="#00FF00"
        app:layout_constraintStart_toEndOf="@id/button4"
        app:layout_constraintTop_toTopOf="@id/button4"
        android:layout_marginStart="15dp" />

    <!-- COLOR BUTTON 3 (BLUE) -->
    <Button
        android:id="@+id/button6"
        android:layout_width="90dp"
        android:layout_height="50dp"
        android:background="#0000FF"
        app:layout_constraintStart_toEndOf="@id/button5"
        app:layout_constraintTop_toTopOf="@id/button5"
        android:layout_marginStart="15dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.experiment2;
mport android.app.Activity;
import android.graphics.Color;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends Activity {
    EditText text1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        text1 = findViewById(R.id.editText1);
```

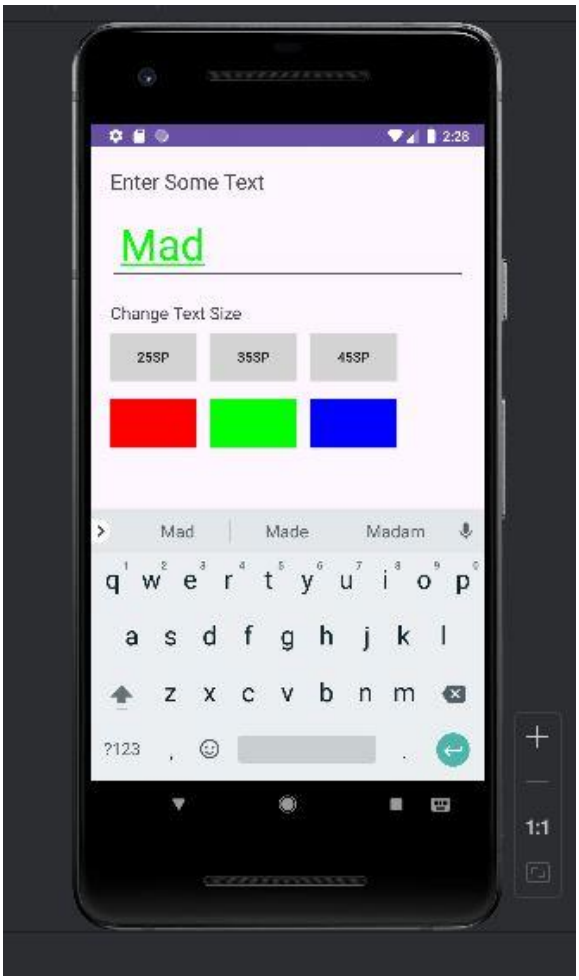
```
        Button size1 = findViewById(R.id.button1);
        Button size2 = findViewById(R.id.button2);
        Button size3 = findViewById(R.id.button3);
        Button col1 = findViewById(R.id.button4);
        Button col2 = findViewById(R.id.button5);
        Button col3 = findViewById(R.id.button6);
        size1.setOnClickListener(v -> text1.setTextSize(25));
        size2.setOnClickListener(v -> text1.setTextSize(35));
        size3.setOnClickListener(v -> text1.setTextSize(45));
        col1.setOnClickListener(v -> text1.setTextColor(Color.RED));
        col2.setOnClickListener(v -> text1.setTextColor(Color.GREEN));
        col3.setOnClickListener(v -> text1.setTextColor(Color.BLUE));
    }
}
```

OUTPUT:

2:27
Enter Some Text
Mad
Change Text Size
25SP
35SP
45SP
Mad
Made
Madam

2:27
Enter Some Text
Mad
Change Text Size
25SP
35SP
45SP
Mad
Made
Madam

2:26
Enter Some Text
Mad
Change Text Size
25SP
35SP
45SP
Mad
Made
Madam
1:1


2:26
Enter Some Text
Mad
Change Text Size
25SP
35SP
45SP
Mad
Made
Madam
1:1

2:28
Enter Some Text
Mad
Change Text Size
25SP
35SP
45SP
Mad
Made
Madam
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123
1:1

Ex.No: 2b

PROGRAM

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="#FFFFFF">

    <TextView
        android:id="@+id/tvTitle"
        android:text="ADDITION"
        android:textStyle="bold"
        android:textSize="22sp"
        android:textColor="#000000"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <!-- Enter No 1 -->
    <TextView
        android:layout_marginTop="20dp"
        android:text="Enter No 1"
        android:textSize="16sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/etNum1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:textSize="18sp" />

    <!-- Enter No 2 -->
    <TextView
        android:layout_marginTop="20dp"
        android:text="Enter No 2"
        android:textSize="16sp"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/etNum2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:textSize="18sp"
        android:backgroundTint="#E91E63" />

    <!-- Buttons Row -->
    <LinearLayout
        android:orientation="horizontal"
        android:layout_marginTop="20dp"
        android:gravity="center"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <Button
            android:id="@+id/btnAdd"
            android:layout_margin="5dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="ADDITION" />

        <Button
            android:id="@+id/btnSub"
            android:layout_margin="5dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="SUBTRACTION" />

        <Button
            android:id="@+id/btnClear"
            android:layout_margin="5dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="CLEAR" />
    </LinearLayout>

    <!-- Output Text -->
    <TextView
```

```
        android:id="@+id/tvAnswer"
        android:layout_marginTop="150dp"
        android:text="ANSWER:"
        android:padding="10dp"
        android:textColor="#FFFFFF"
        android:background="#777777"
        android:textSize="17sp"
        android:gravity="center"
        android:layout_gravity="center_horizontal"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:paddingLeft="20dp"
        android:paddingRight="20dp"
        android:backgroundTint="#777777" />

</LinearLayout>
```

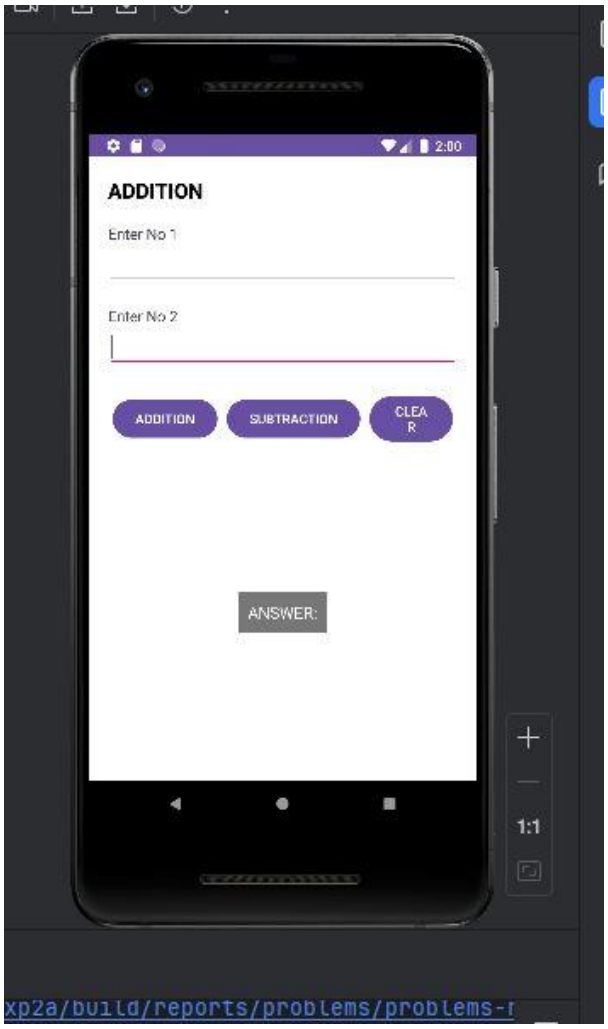
MainActivity.java

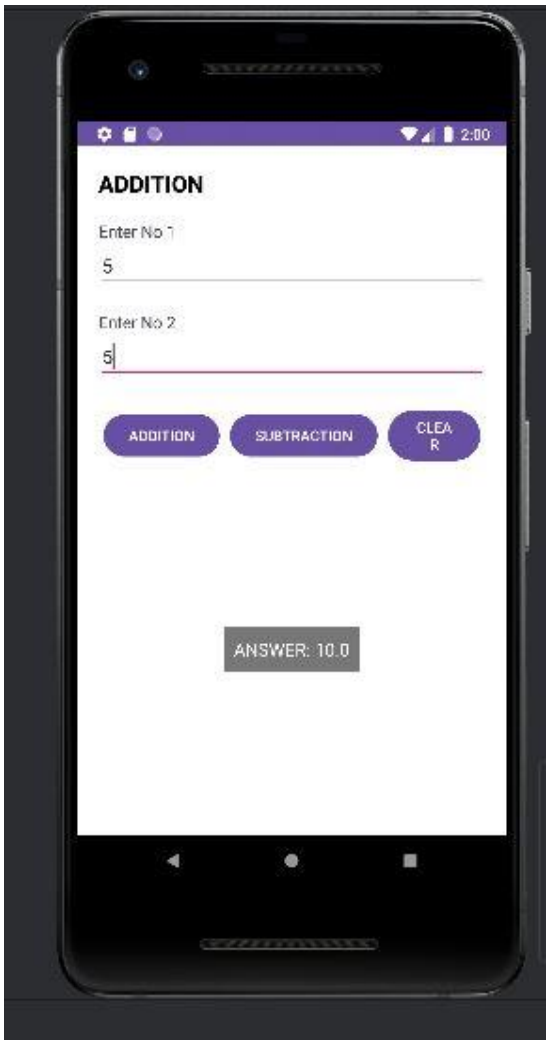
```
package com.example.exp2b;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText etNum1, etNum2;
    Button btnAdd, btnSub, btnClear;
    TextView tvAnswer;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etNum1 = findViewById(R.id.etNum1);
        etNum2 = findViewById(R.id.etNum2);
        btnAdd = findViewById(R.id.btnAdd);
```

```
        btnSub = findViewById(R.id.btnSub);
        btnClear = findViewById(R.id.btnClear);
        tvAnswer = findViewById(R.id.tvAnswer);

        btnAdd.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                calculate('+');
            }
        });

        btnSub.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                calculate('-');
            }
        });

        btnClear.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                etNum1.setText("");
                etNum2.setText("");
                tvAnswer.setText("ANSWER:");
            }
        });
    }
    private void calculate(char type) {

        try {
            double n1 = Double.parseDouble(etNum1.getText().toString());
            double n2 = Double.parseDouble(etNum2.getText().toString());
            double ans = 0;

            if (type == '+')
                ans = n1 + n2;
            else if (type == '-')
                ans = n1 - n2;

            tvAnswer.setText("ANSWER: " + ans);

        } catch (Exception e) {
```

```
        Toast.makeText(MainActivity.this, "Please enter valid numbers", Toast.LENGTH_SHORT).show();
    }
  }}
```

OUTPUT





EX.NO: 3

PROGRAM

activity_main.xml

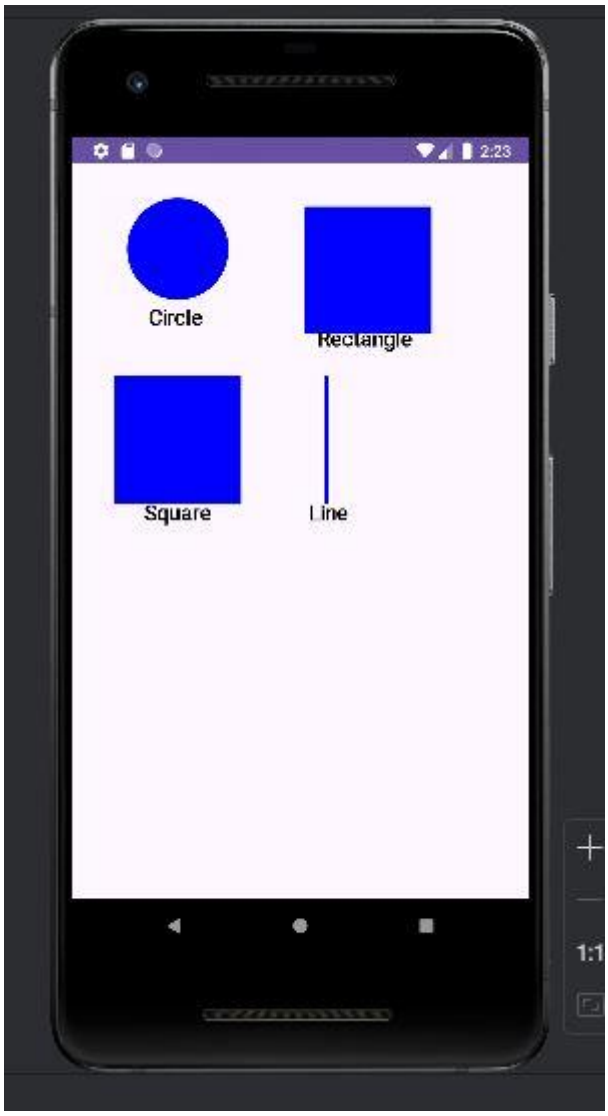
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <!-- Shapes will be drawn from Java code -->
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.exp3;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Context;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.os.Bundle;
import android.view.View;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Load custom view that draws shapes + text
        setContentView(new ShapeView(this));
    }
}

// ------------------------------------------------------------------
// CUSTOM VIEW CLASS (SHAPES + TEXT BELOW EACH SHAPE) — SINGLE FILE
// ------------------------------------------------------------------
class ShapeView extends View {
```

```
    Paint shapePaint, textPaint;
    public ShapeView(Context context) {
        super(context);
        shapePaint = new Paint();
        shapePaint.setColor(Color.BLUE);
        shapePaint.setStrokeWidth(10);
        shapePaint.setAntiAlias(true);
        textPaint = new Paint();
        textPaint.setColor(Color.BLACK);
        textPaint.setTextSize(50f);
        textPaint.setFakeBoldText(true);
    }
    @Override
    protected void onDraw(Canvas canvas) {
        super.onDraw(canvas);

        // Positions
        int cx = 250;  // Circle center X
        int cy = 200;  // Circle center Y

        // ------------------ CIRCLE ------------------
        canvas.drawCircle(cx, cy, 120, shapePaint);
        canvas.drawText("Circle", cx - 70, cy + 180, textPaint);

        // ---------------- RECTANGLE -----------------
        canvas.drawRect(550, 100, 850, 400, shapePaint);
        canvas.drawText("Rectangle", 580, 430, textPaint);

        // ------------------ SQUARE -------------------
        canvas.drawRect(100, 500, 400, 800, shapePaint);
        canvas.drawText("Square", 170, 840, textPaint);

        // ------------------ LINE ---------------------
        canvas.drawLine(600, 500, 600, 800, shapePaint);
        canvas.drawText("Line", 560, 840, textPaint);
    }
}
```

OUTPUT:



EX.NO: 4

PROGRAM

activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Student Details"
            android:textSize="24sp"
            android:textStyle="bold"
            android:gravity="center"
            android:padding="10dp"/>

        <EditText
            android:id="@+id/editRoll"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Roll No"
            android:inputType="number"/>

        <EditText
            android:id="@+id/editName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Name"/>

        <EditText
            android:id="@+id/editMarks"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Marks"
```

```
        android:inputType="number"/>

    <Button
        android:id="@+id/btnInsert"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="INSERT"/>

    <Button
        android:id="@+id/btnUpdate"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="UPDATE"/>

    <Button
        android:id="@+id/btnDelete"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="DELETE"/>

    <Button
        android:id="@+id/btnView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="VIEW"/>

    <Button
        android:id="@+id/btnViewAll"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="VIEW ALL"/>

  </LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.example.experiment4;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.content.ContentValues;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
```

```
import android.os.Bundle;
import android.text.TextUtils;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity {

    SQLiteDatabase db;
    EditText editRoll, editName, editMarks;
    Button btnInsert, btnUpdate, btnDelete, btnView, btnViewAll;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editRoll = findViewById(R.id.editRoll);
        editName = findViewById(R.id.editName);
        editMarks = findViewById(R.id.editMarks);

        btnInsert = findViewById(R.id.btnInsert);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);
        btnView = findViewById(R.id.btnView);
        btnViewAll = findViewById(R.id.btnViewAll);

        db = openOrCreateDatabase("StudentDB", MODE_PRIVATE, null);

        db.execSQL("CREATE TABLE IF NOT EXISTS student(" +
                "ROLLNO INTEGER PRIMARY KEY, " +
                "NAME TEXT, " +
                "MARKS INTEGER)");

        // INSERT
        btnInsert.setOnClickListener(v -> {

            if (!validateAllFields()) return;

            ContentValues cv = new ContentValues();
            cv.put("ROLLNO", Integer.parseInt(editRoll.getText().toString()));
            cv.put("NAME", editName.getText().toString());
            cv.put("MARKS", Integer.parseInt(editMarks.getText().toString()));

            long result = db.insert("student", null, cv);
```

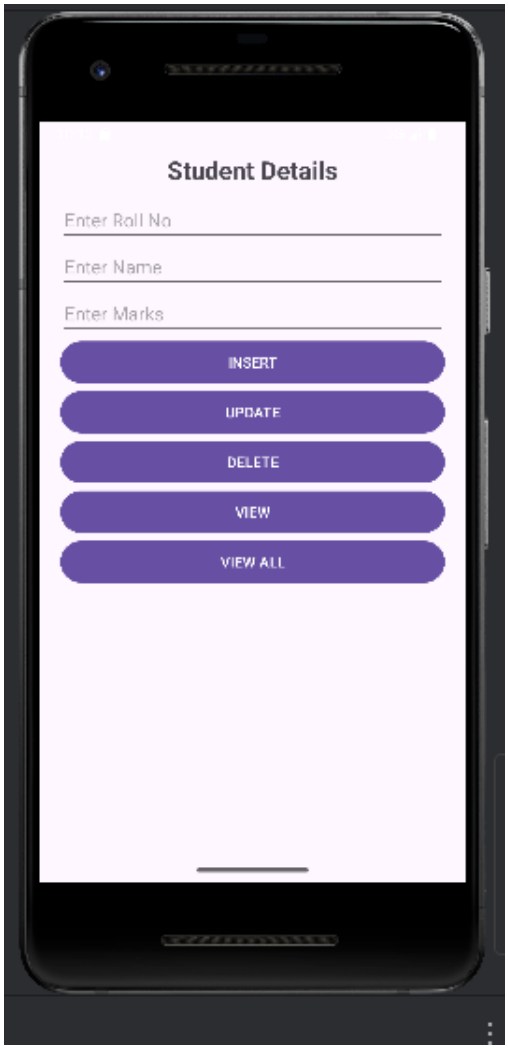
```
        if (result == -1)
            showMessage("Error", "Roll No Already Exists");
        else {
            showMessage("Success", "Record Inserted");
            clearFields();
        }
    });

    // UPDATE
    btnUpdate.setOnClickListener(v -> {

        if (!validateAllFields()) return;

        ContentValues cv = new ContentValues();
        cv.put("NAME", editName.getText().toString());
        cv.put("MARKS", Integer.parseInt(editMarks.getText().toString()));

        int rows = db.update("student",
                cv,
                "ROLLNO=?",
                new String[]{editRoll.getText().toString()});

        if (rows > 0) {
            showMessage("Success", "Record Updated");
            clearFields();
        } else
            showMessage("Error", "Record Not Found");
    });

    // DELETE
    btnDelete.setOnClickListener(v -> {

        if (TextUtils.isEmpty(editRoll.getText().toString())) {
            showMessage("Error", "Enter Roll No");
            return;
        }

        int rows = db.delete("student",
                "ROLLNO=?",
                new String[]{editRoll.getText().toString()});

        if (rows > 0) {
```

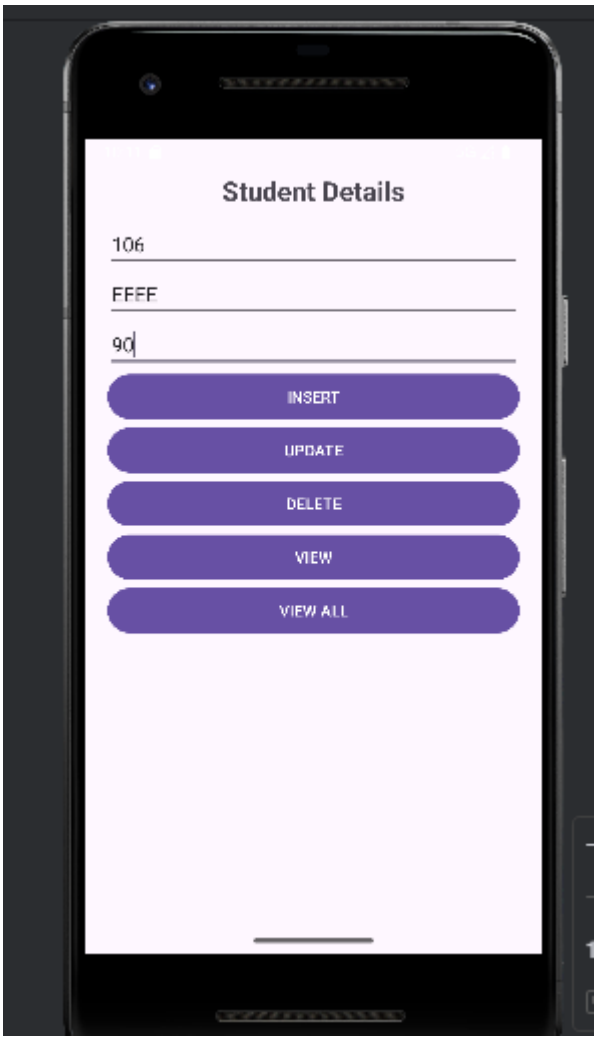
```
            showMessage("Success", "Record Deleted");
            clearFields();
        } else
            showMessage("Error", "Record Not Found");
    });

    // VIEW
    btnView.setOnClickListener(v -> {

        if (TextUtils.isEmpty(editRoll.getText().toString())) {
            showMessage("Error", "Enter Roll No");
            return;
        }

        Cursor c = db.rawQuery("SELECT * FROM student WHERE
ROLLNO=?",
                new String[]{editRoll.getText().toString()});

        if (c.moveToFirst()) {
            editName.setText(c.getString(1));
            editMarks.setText(c.getString(2));
            showMessage("Success", "Record Found");
        } else
            showMessage("Error", "Record Not Found");

        c.close();
    });

    // VIEW ALL
    btnViewAll.setOnClickListener(v -> {

        Cursor c = db.rawQuery("SELECT * FROM student", null);

        if (c.getCount() == 0) {
            showMessage("Error", "No Records Found");
            return;
        }

        StringBuilder buffer = new StringBuilder();

        while (c.moveToNext()) {
            buffer.append("Roll No: ").append(c.getInt(0)).append("\n");
            buffer.append("Name: ").append(c.getString(1)).append("\n");
```

```
            buffer.append("Marks: ").append(c.getInt(2)).append("\n\n");
        }

        showMessage("All Student Records", buffer.toString());
        c.close();
    });
}

private boolean validateAllFields() {

    if (TextUtils.isEmpty(editRoll.getText().toString()) ||
            TextUtils.isEmpty(editName.getText().toString()) ||
            TextUtils.isEmpty(editMarks.getText().toString())) {

        showMessage("Error", "All Fields Required");
        return false;
    }
    return true;
}

private void showMessage(String title, String message) {
    AlertDialog.Builder builder = new AlertDialog.Builder(this);
    builder.setTitle(title);
    builder.setMessage(message);
    builder.setCancelable(true);
    builder.show();
}

private void clearFields() {
    editRoll.setText("");
    editName.setText("");
    editMarks.setText("");
}
}
```

OUTPUT

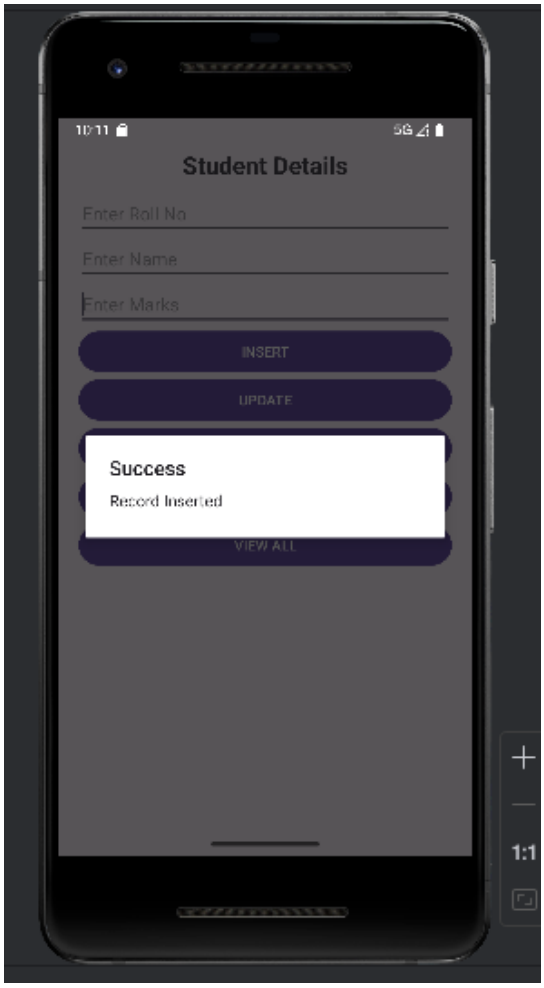
INSERT RECORD

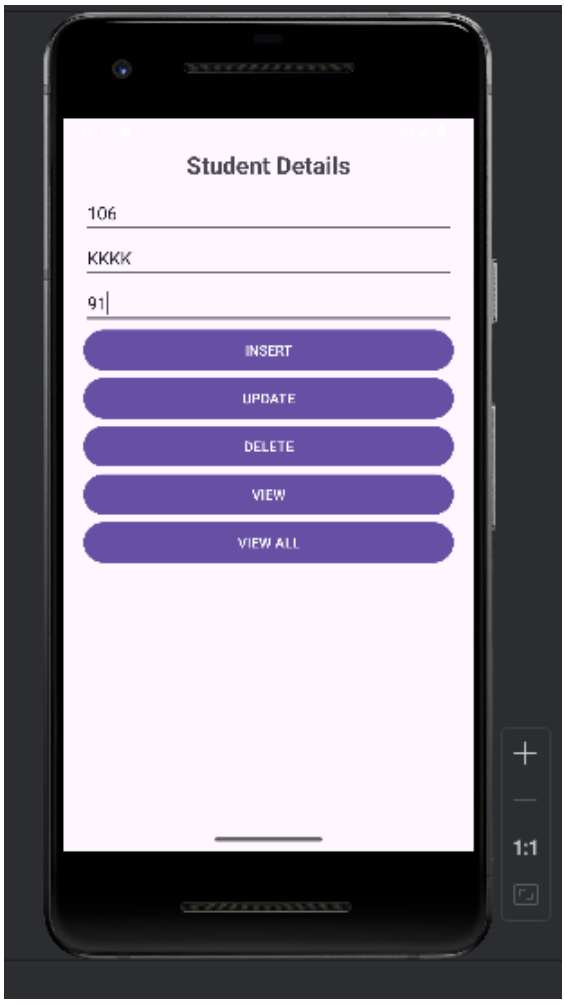




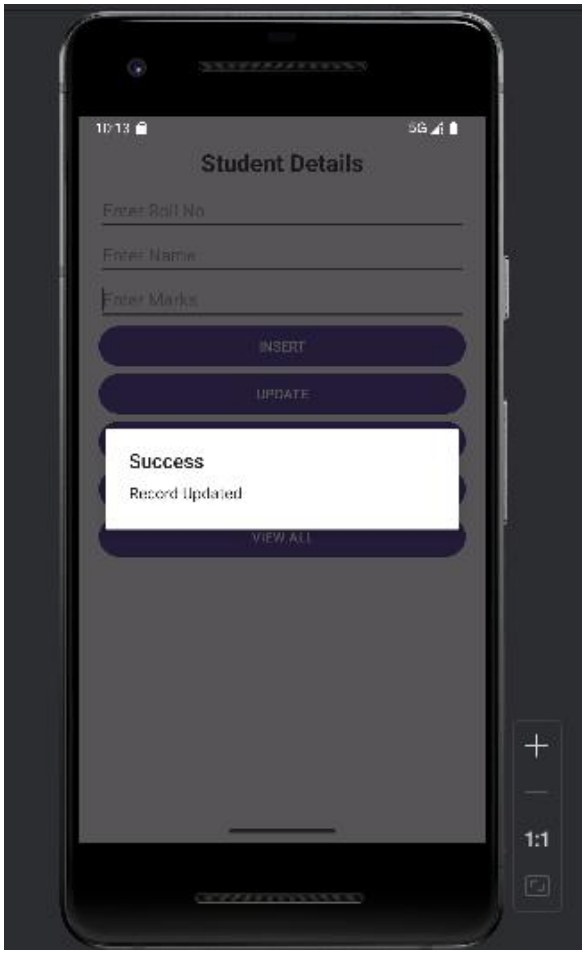
UPDATE

AFTER INSERT

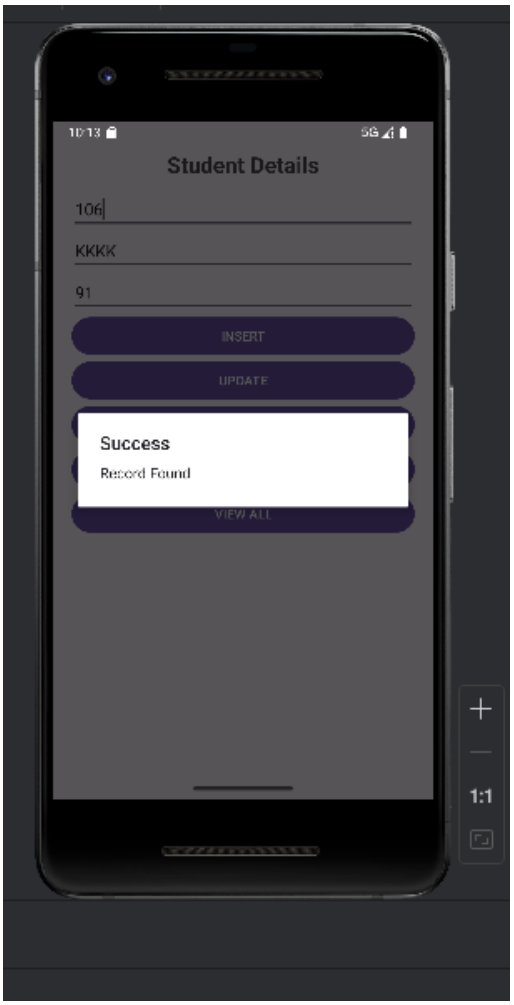




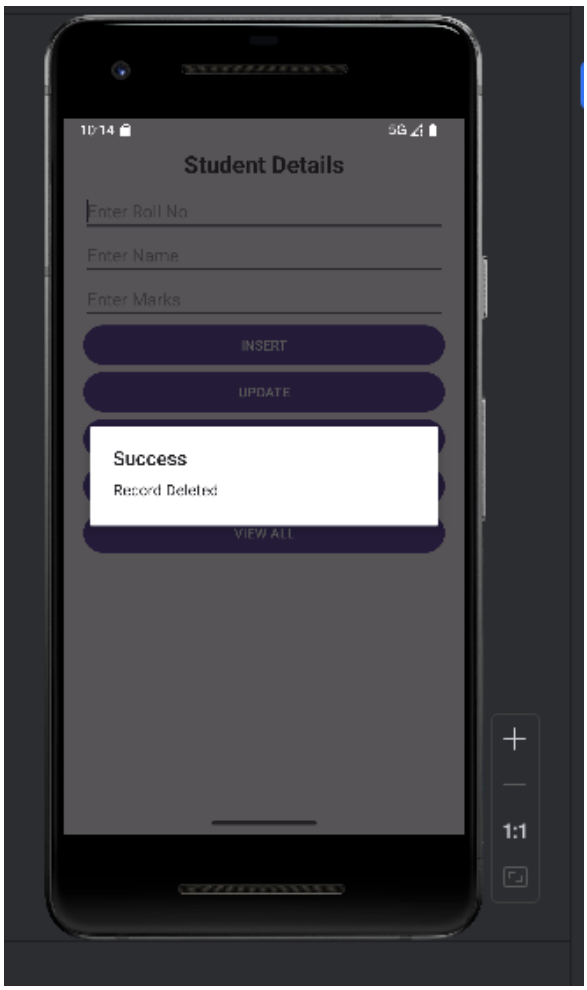
AFTER UPDATE



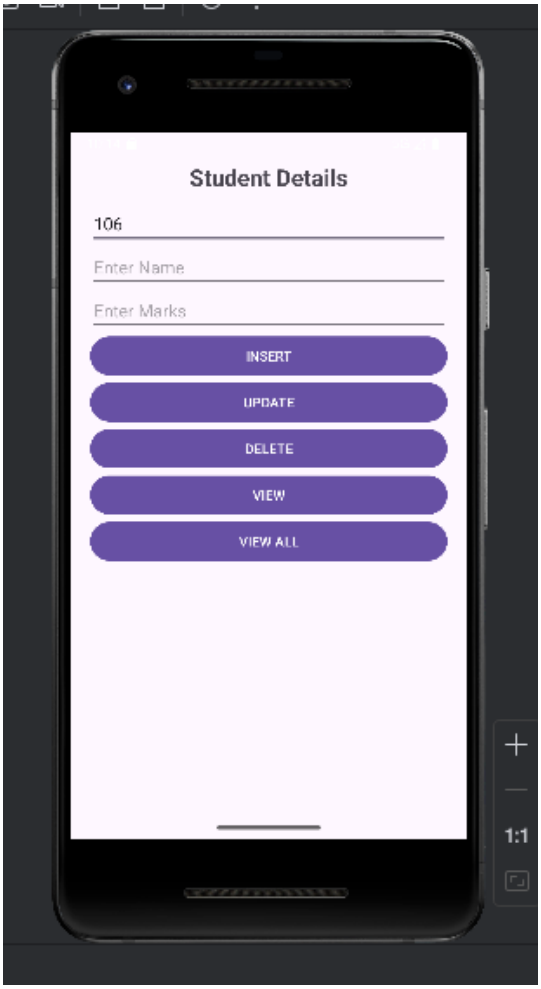
VIEW (FOUND)



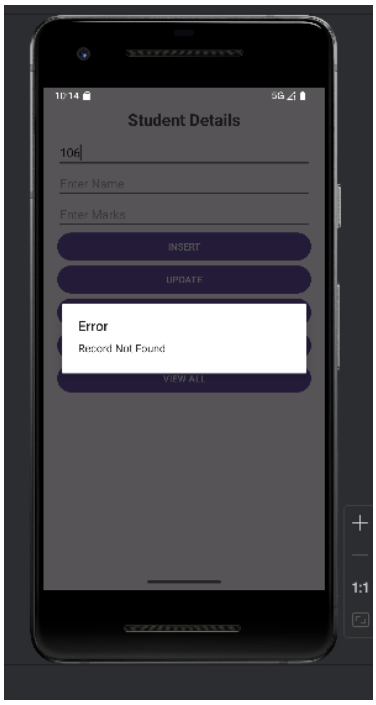
DELETION (RECORD NOT FOUND)



AFTER DELETION



VIEW (NOT FOUIND)



STUDENT DATABASE BEFORE OPERATIONS

student

Live updates

	ROLLNO	NAME	MARKS
1	102	AAA	95
2	103	BBBB	100
3	104	CCCC	98

STUDENT DATABASE AFTER OPERATIONS

student

Live updates

	ROLLNO	NAME	MARKS
1	102	AAA	95
2	103	BBBB	100
3	104	CCCC	98
4	105	JJJJ	94

EX.NO: 5

PROGRAM

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   package="com.example.experiment5">
   <uses-permission
android:name="android.permission.POST_NOTIFICATIONS"/>
   <application
      android:allowBackup="true"
      android:theme="@style/Theme.AppCompat.Light.DarkActionBar"
      android:label="My Application">
      <activity
         android:name=".MainActivity"
         android:exported="true">
         <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
         </intent-filter>
      </activity>
   </application> </manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <TextView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="MESSAGE FROM SACET"
      android:textSize="18sp"
      android:padding="15dp"
      android:background="#E0E0E0"
      android:textAlignment="center"/>
   <Button
```

```
        android:id="@+id/btnNotify"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="NOTIFY"
        android:layout_marginTop="40dp"/>
</LinearLayout>
```

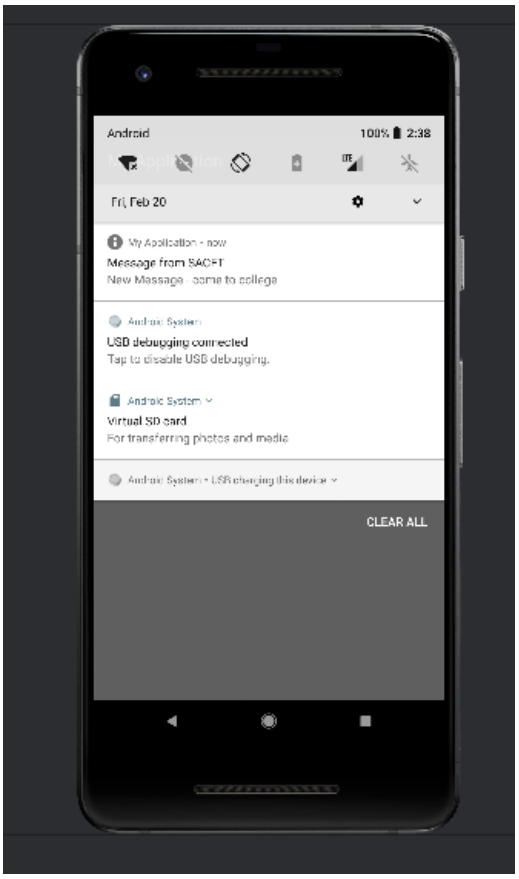
MainActivity.java

```
package com.example.experiment5;
import android.Manifest;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
public class MainActivity extends AppCompatActivity {
    private static final String CHANNEL_ID = "SACET_CHANNEL";
    Button btnNotify;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnNotify = findViewById(R.id.btnNotify);

        createNotificationChannel();

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU)
{
            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.POST_NOTIFICATIONS},
                    101);
        }

        btnNotify.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                showNotification();
            }
```

```
        });
    }
    private void showNotification() {
        NotificationCompat.Builder builder =
                new NotificationCompat.Builder(this, CHANNEL_ID)
                        .setSmallIcon(android.R.drawable.ic_dialog_info)
                        .setContentTitle("Message from SACET")
                        .setContentText("New Message - come to college")
                        .setPriority(NotificationCompat.PRIORITY_DEFAULT);
        NotificationManagerCompat notificationManager =
                NotificationManagerCompat.from(this);
        notificationManager.notify(1, builder.build());
    }
    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

            CharSequence name = "SACET Channel";
            String description = "Channel for SACET Notifications";
            int importance = NotificationManager.IMPORTANCE_DEFAULT;

            NotificationChannel channel =
                    new NotificationChannel(CHANNEL_ID, name, importance);
            channel.setDescription(description);
            NotificationManager notificationManager =
                    getSystemService(NotificationManager.class);
            notificationManager.createNotificationChannel(channel);
        }  }}
```

OUTPUT

2:37
My Application
MESSAGE FROM SACET
NOTIFY


Android
100% 2:38
Fri, Feb 20
My Application • now
Message from SACET
New Message : come to college
Android System
USB debugging connected
Tap to disable USB debugging.
Android System
Virtual SD card
For transferring photos and media
CLEAR ALL

EX.NO: 6

PROGRAM

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.multithreading">

    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar"
        android:label="Multithreading App">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>
    </application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="250dp"
        android:layout_height="200dp"
        android:src="@drawable/sample1"
        android:scaleType="centerCrop"
```

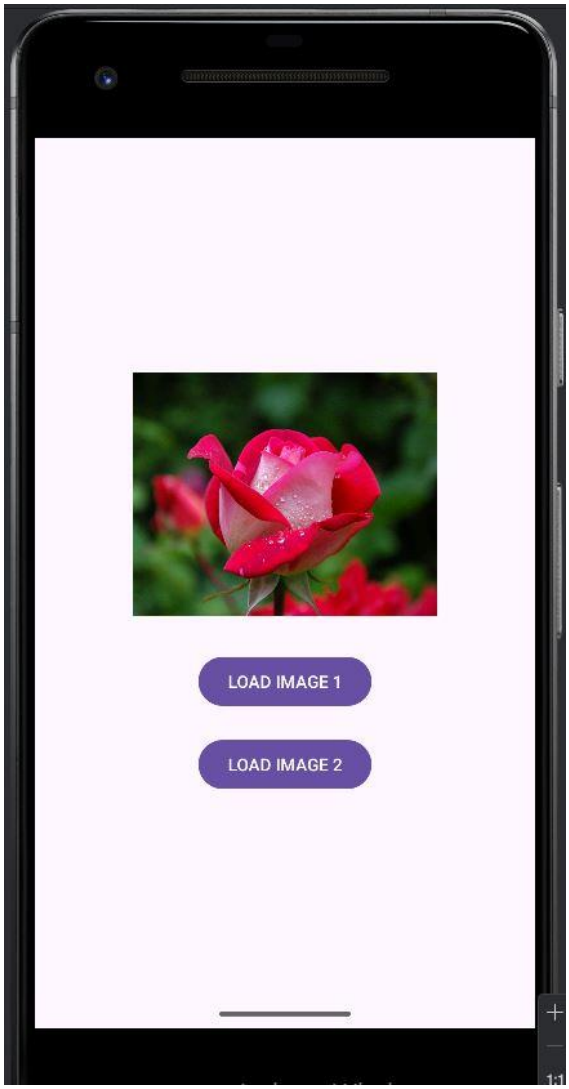
```
        android:layout_marginBottom="30dp"/>

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="LOAD IMAGE 1"/>

    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="LOAD IMAGE 2"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

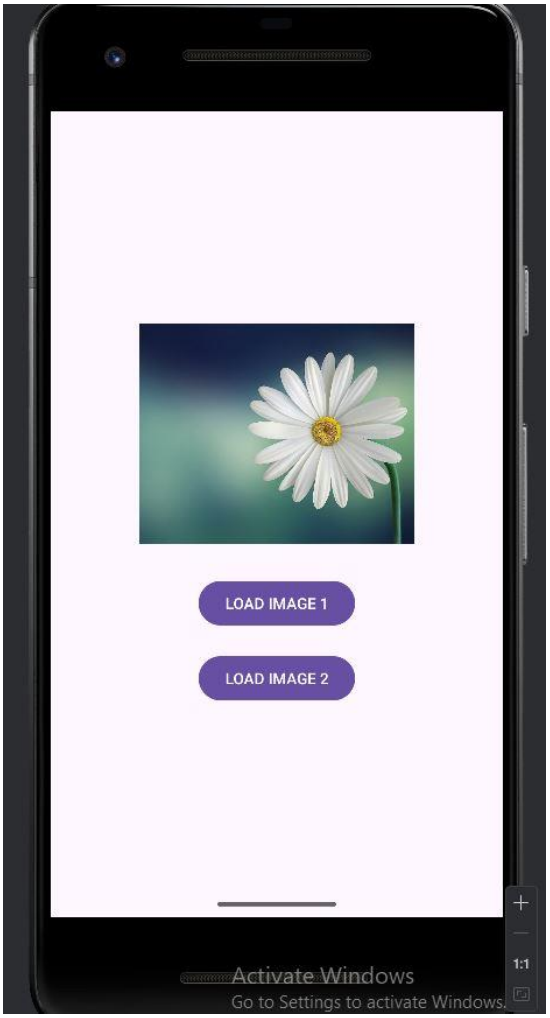
MainActivity.java

```
package com.example.multithreading;

import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.Button;
import android.widget.ImageView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ImageView imageView;
    Button btn1, btn2;

    Handler handler = new Handler(Looper.getMainLooper());

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        imageView = findViewById(R.id.imageView);
        btn1 = findViewById(R.id.btn1);
```

```
        btn2 = findViewById(R.id.btn2);

        btn1.setOnClickListener(v -> {
            new Thread(() -> {
                try {
                    Thread.sleep(2000); // simulate delay
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }

                handler.post(() ->
imageView.setImageResource(R.drawable.sample1));
            }).start();
        });

        btn2.setOnClickListener(v -> {
            new Thread(() -> {
                try {
                    Thread.sleep(2000);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }

                handler.post(() ->
imageView.setImageResource(R.drawable.sample2));
            }).start();
        });
    }
}
```

OUTPUT


LOAD IMAGE 1
LOAD IMAGE 2


LOAD IMAGE 1
LOAD IMAGE 2
Activate Windows
Go to Settings to activate Windows.

EX.NO: 7

PROGRAM

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
    <uses-permission
android:name="android.permission.ACCESS_FINE_LOCATION"/>
    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar"
        android:label="Location App">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
```

```
    android:gravity="center"
    android:padding="20dp">
    <TextView
        android:id="@+id/tvLatitude"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Latitude:"
        android:textSize="18sp"
        android:layout_marginBottom="10dp"/>

    <TextView
        android:id="@+id/tvLongitude"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Longitude:"
        android:textSize="18sp"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnLocation"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="GET LOCATION"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.ex7;
import android.Manifest;
```

```
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
public class MainActivity extends AppCompatActivity {
    TextView tvLatitude, tvLongitude;
    Button btnLocation;
    LocationManager locationManager;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        tvLatitude = findViewById(R.id.tvLatitude);
        tvLongitude = findViewById(R.id.tvLongitude);
        btnLocation = findViewById(R.id.btnLocation);
        locationManager = (LocationManager)
getSystemService(LOCATION_SERVICE);
        btnLocation.setOnClickListener(v -> getLocation());
    }
    private void getLocation() {
        if (ActivityCompat.checkSelfPermission(this,
                Manifest.permission.ACCESS_FINE_LOCATION) !=
PackageManager.PERMISSION_GRANTED) {
```

```
            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.ACCESS_FINE_LOCATION},
101);
            return;
        }
        try {
            Location location =
locationManager.getLastKnownLocation(LocationManager.GPS_PROVIDER);
            if (location != null) {
                tvLatitude.setText("Latitude: " + location.getLatitude());
                tvLongitude.setText("Longitude: " + location.getLongitude());
            } else {
                // Default for emulator
                tvLatitude.setText("Latitude: 13.0827");
                tvLongitude.setText("Longitude: 80.2707");
            }
        } catch (Exception e) {
            tvLatitude.setText("Latitude: 13.0827");
            tvLongitude.setText("Longitude: 80.2707");
        }
    }
}
```

OUTPUT:

Latitude: 37.606181666666664
Longitude: -122.21719
GET LOCATION

EX.NO:8

PROGRAM

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.experiment8">

    <uses-permission
android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
    <uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE"/>

    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar"
        android:label="SD Card App"
        android:requestLegacyExternalStorage="true">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>

        </activity>
    </application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">
```

```
    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text"
        android:textSize="18sp"/>

    <Button
        android:id="@+id/btnWrite"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="WRITE DATA"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnRead"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="READ DATA"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/btnClear"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="CLEAR"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

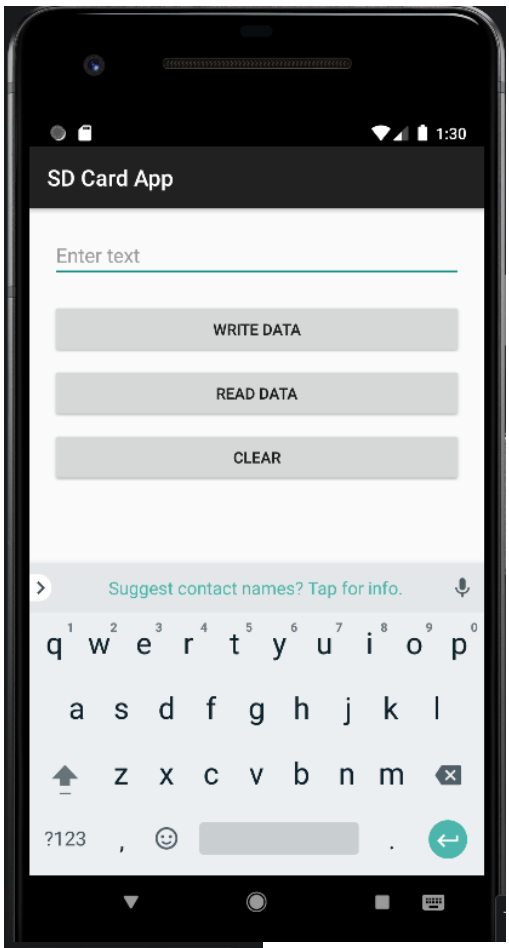
MainActivity.java

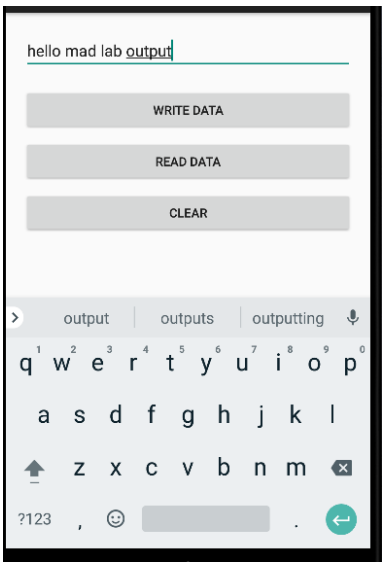
```
package com.example.experiment8;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.os.Environment;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;

public class MainActivity extends AppCompatActivity {

    EditText editText;
    Button btnWrite, btnRead, btnClear;
    File file;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editText = findViewById(R.id.editText);
        btnWrite = findViewById(R.id.btnWrite);
        btnRead = findViewById(R.id.btnRead);
        btnClear = findViewById(R.id.btnClear);

        // Permission
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.M) {
            if (ContextCompat.checkSelfPermission(this,
                    Manifest.permission.WRITE_EXTERNAL_STORAGE)
                    != PackageManager.PERMISSION_GRANTED) {

                ActivityCompat.requestPermissions(this,
                        new
String[]{Manifest.permission.WRITE_EXTERNAL_STORAGE,
                            Manifest.permission.READ_EXTERNAL_STORAGE}, 1);
            }
        }

        file = new File(Environment.getExternalStorageDirectory(), "myfile.txt");

        // WRITE
        btnWrite.setOnClickListener(v -> {
            try {
```

```
            FileOutputStream fos = new FileOutputStream(file);
            fos.write(editText.getText().toString().getBytes());
            fos.close();

            Toast.makeText(this,
                    "Data Written in SDCARD",
                    Toast.LENGTH_LONG).show();

        } catch (Exception e) {
            e.printStackTrace();
        }
    });

    // READ
    btnRead.setOnClickListener(v -> {
        try {
            FileInputStream fis = new FileInputStream(file);
            int i;
            String data = "";

            while ((i = fis.read()) != -1) {
                data += (char) i;
            }

            editText.setText(data);

            Toast.makeText(this,
                    "Data Read from SDCARD",
                    Toast.LENGTH_SHORT).show();

        } catch (Exception e) {
            e.printStackTrace();
        }
    });

    // CLEAR
    btnClear.setOnClickListener(v -> editText.setText(""));
  }
}
```

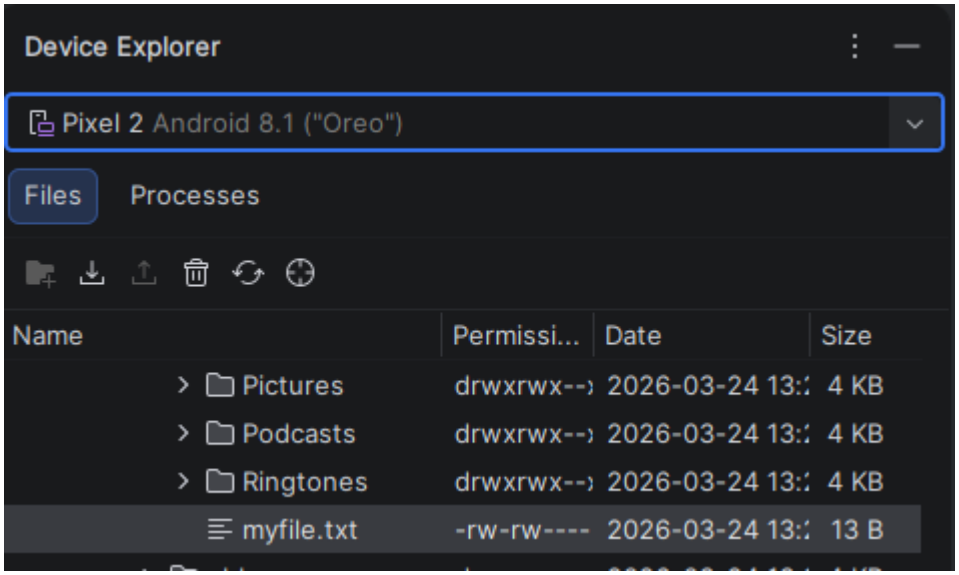
OUTPUT

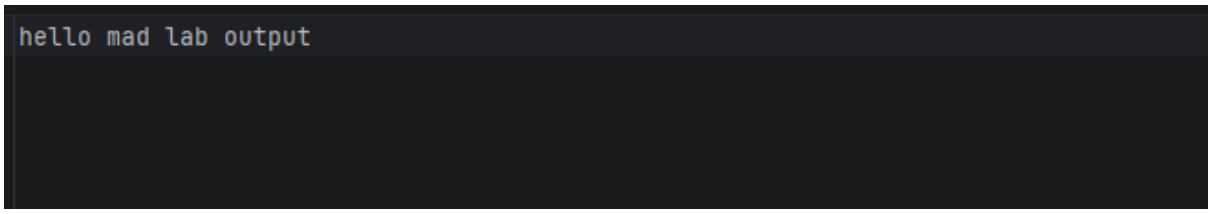

1:30
SD Card App
Enter text
WRITE DATA
READ DATA
CLEAR
Suggest contact names? Tap for info.


hello mad lab output
WRITE DATA
READ DATA
CLEAR
output
outputs
outputting

SD Card App

SD Card App
hello mad lab output
WRITE DATA
READ DATA
CLEAR


Device Explorer
Pixel 2 Android 8.1 ("Oreo")
Files
Processes
Name
Permissi...
Date
Size
Pictures
drwxrwx--
2026-03-24 13:
4 KB
Podcasts
drwxrwx--
2026-03-24 13:
4 KB
Ringtones
drwxrwx--
2026-03-24 13:
4 KB
myfile.txt
-rw-rw----
2026-03-24 13:
13 B


hello mad lab output

EX. NO:9

PROGRAM

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.smsapp">

    <uses-permission android:name="android.permission.SEND_SMS"/>

    <application
        android:allowBackup="true"
        android:label="SMS Application">

        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>

    </application>

</manifest>
```

activity_main.xml

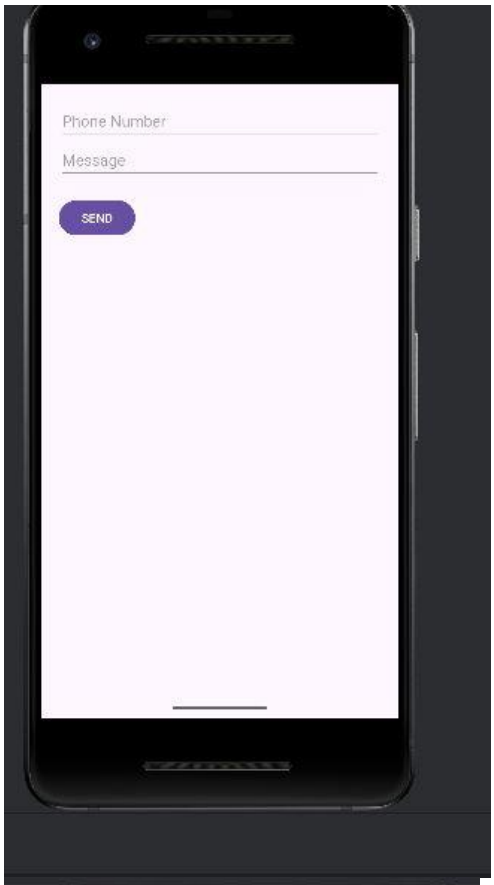
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/phone"
        android:hint="Phone Number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/message"
        android:hint="Message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

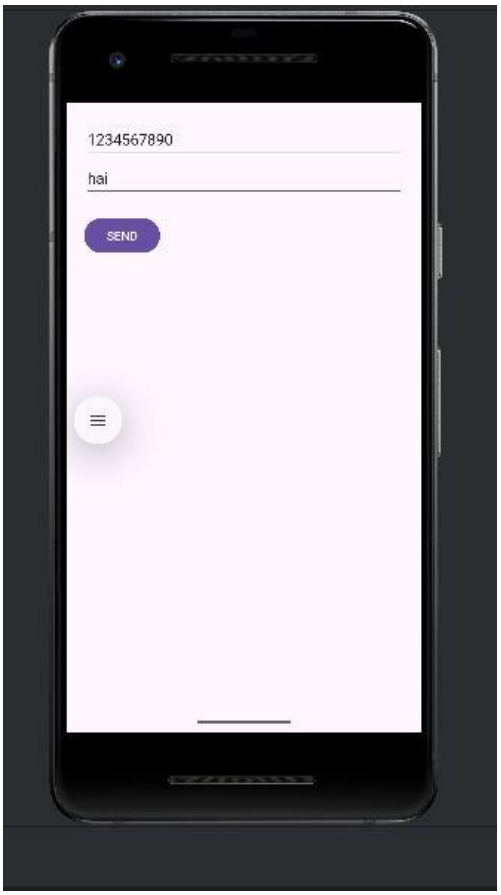
```
    <Button
        android:id="@+id/sendSMS"
        android:text="SEND"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

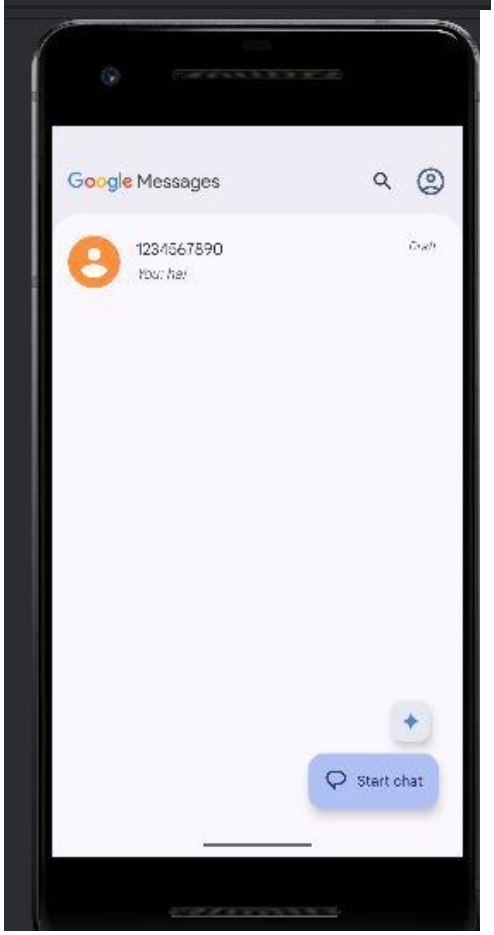
MainActivity.java

```
package com.example.smsapp;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {

    EditText phone, message;
    Button send;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        phone = findViewById(R.id.phone);
        message = findViewById(R.id.message);
        send = findViewById(R.id.sendSMS);

        send.setOnClickListener(v -> {

            Intent intent = new Intent(Intent.ACTION_SENDTO);
            intent.setData(Uri.parse("smsto:" + phone.getText().toString()));
            intent.putExtra("sms_body", message.getText().toString());

            startActivity(intent);
        });
    }
}
```

OUTPUT


Phone Number
Message
SEND


1234567890
hai
SEND


Google Messages
1234567890
Start chat

EX NO: 10

PROGRAM

activity_main.xml

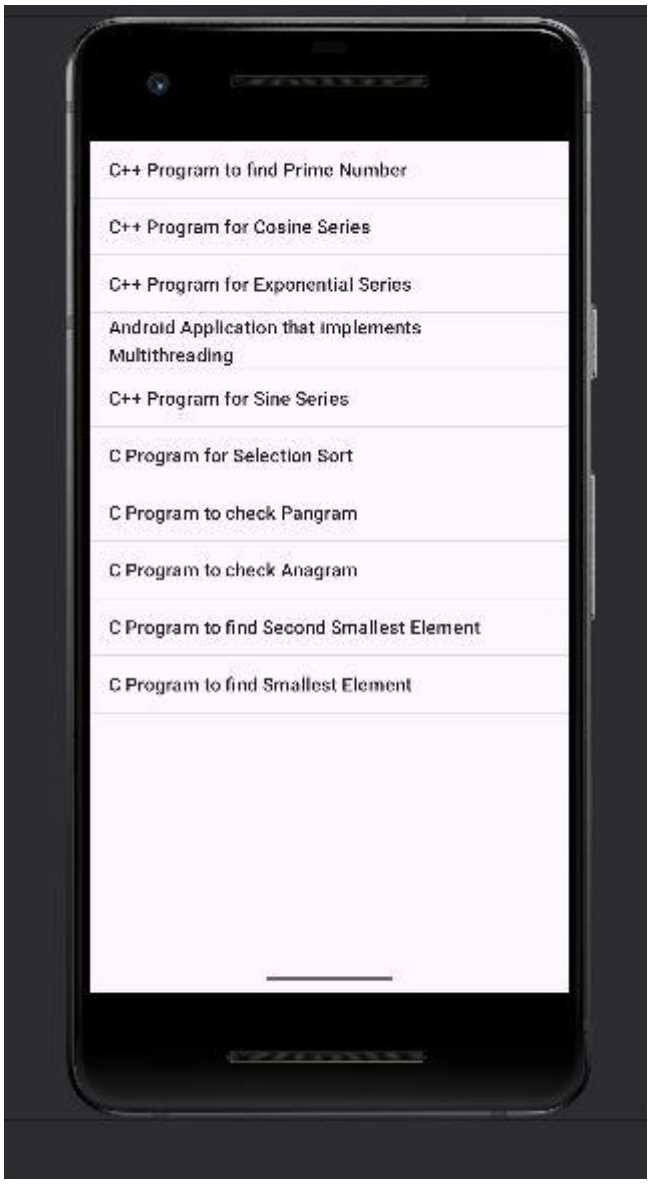
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.listviewapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;

public class MainActivity extends AppCompatActivity {

    ListView listView;

    String[] programs = {
            "C++ Program to find Prime Number",
            "C++ Program for Cosine Series",
            "C++ Program for Exponential Series",
            "Android Application that implements Multithreading",
            "C++ Program for Sine Series",
            "C Program for Selection Sort",
            "C Program to check Pangram",
            "C Program to check Anagram",
            "C Program to find Second Smallest Element",
            "C Program to find Smallest Element"
    };

    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        ArrayAdapter<String> adapter =
                new ArrayAdapter<>(this,
                        android.R.layout.simple_list_item_1,
                        programs);

        listView.setAdapter(adapter);
    }
}
```

OUTPUT


C++ Program to find Prime Number
C++ Program for Cosine Series
C++ Program for Exponential Series
Android Application that implements Multithreading
C++ Program for Sine Series
C Program for Selection Sort
C Program to check Pangram
C Program to check Anagram
C Program to find Second Smallest Element
C Program to find Smallest Element

EX. NO:11

PROGRAM

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.exp9">

    <application
        android:allowBackup="true"
        android:label="Email Application"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>

    </application>

</manifest>
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etTo"
        android:hint="Recipient"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etSubject"
        android:hint="Subject"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

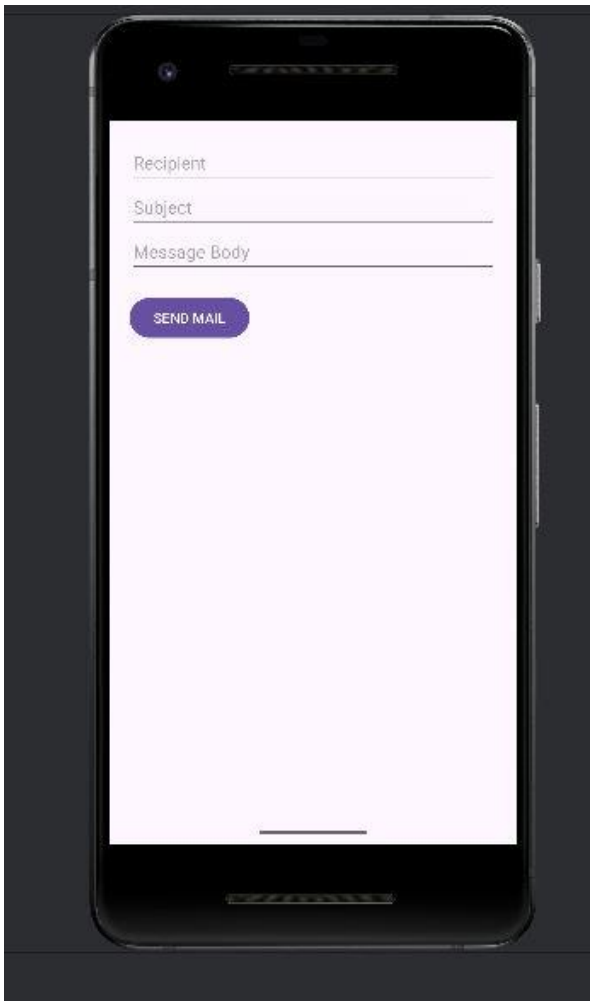
```
    <EditText
        android:id="@+id/etMessage"
        android:hint="Message Body"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnSend"
        android:text="SEND MAIL"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

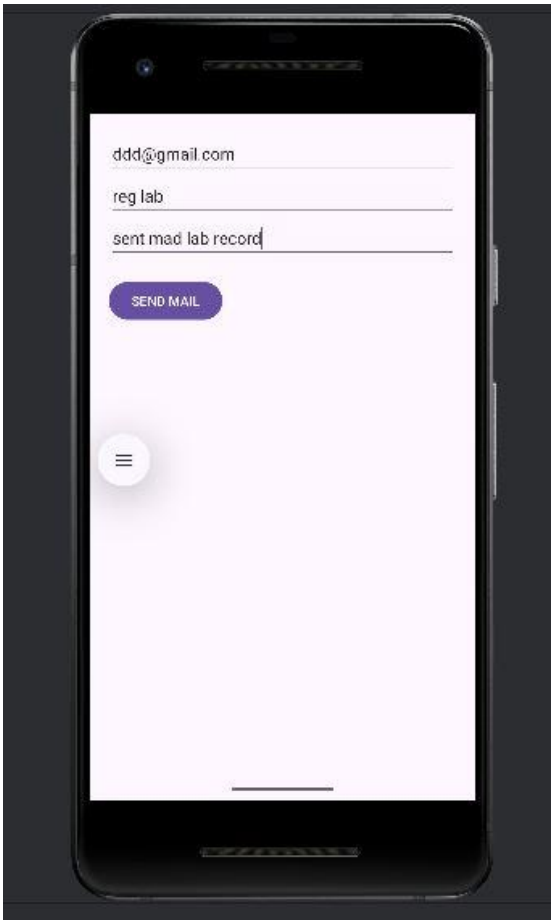
MainActivity.java

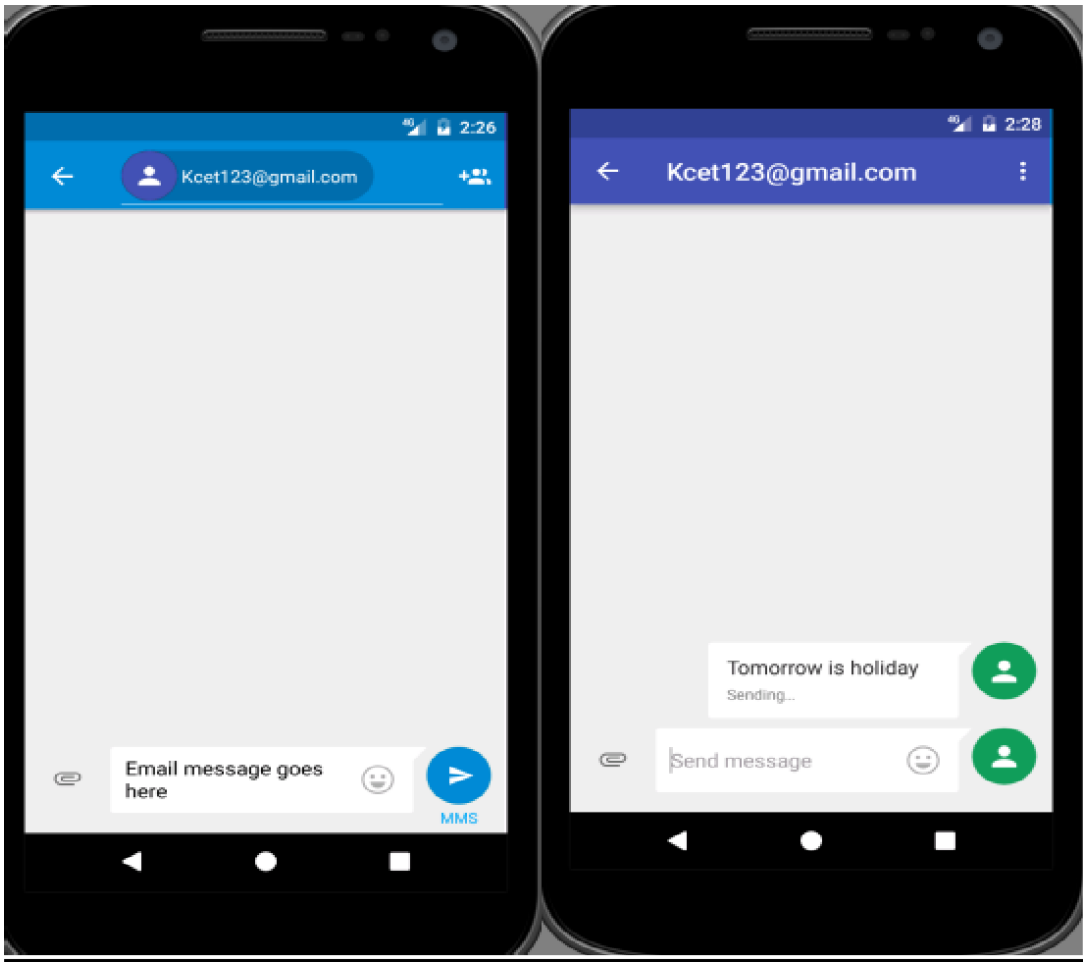
```
package com.example.exp9;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText to, subject, message;
    Button send;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        to = findViewById(R.id.etTo);
        subject = findViewById(R.id.etSubject);
        message = findViewById(R.id.etMessage);
        send = findViewById(R.id.btnSend);

        send.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
            public void onClick(View v) {

                Intent intent = new Intent(Intent.ACTION_SEND);
                intent.setType("message/rfc822");

                intent.putExtra(Intent.EXTRA_EMAIL,
                        new String[]{to.getText().toString()});

                intent.putExtra(Intent.EXTRA_SUBJECT,
                        subject.getText().toString());

                intent.putExtra(Intent.EXTRA_TEXT,
                        message.getText().toString());

                startActivity(Intent.createChooser(intent,"Send Email"));
            }
        });
    }
}
```

OUTPUT


Recipient
Subject
Message Body
SEND MAIL


ddd@gmail.com
reg lab
sent mad lab record
SEND MAIL


2:26
Kcet123@gmail.com
Email message goes here
MMS
2:28
Kcet123@gmail.com
Tomorrow is holiday
Sending...
Send message

MINI –PROJECT